



NOAA
FISHERIES

Integration of Fourier transform near-infrared spectroscopy into the production ageing process

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USA

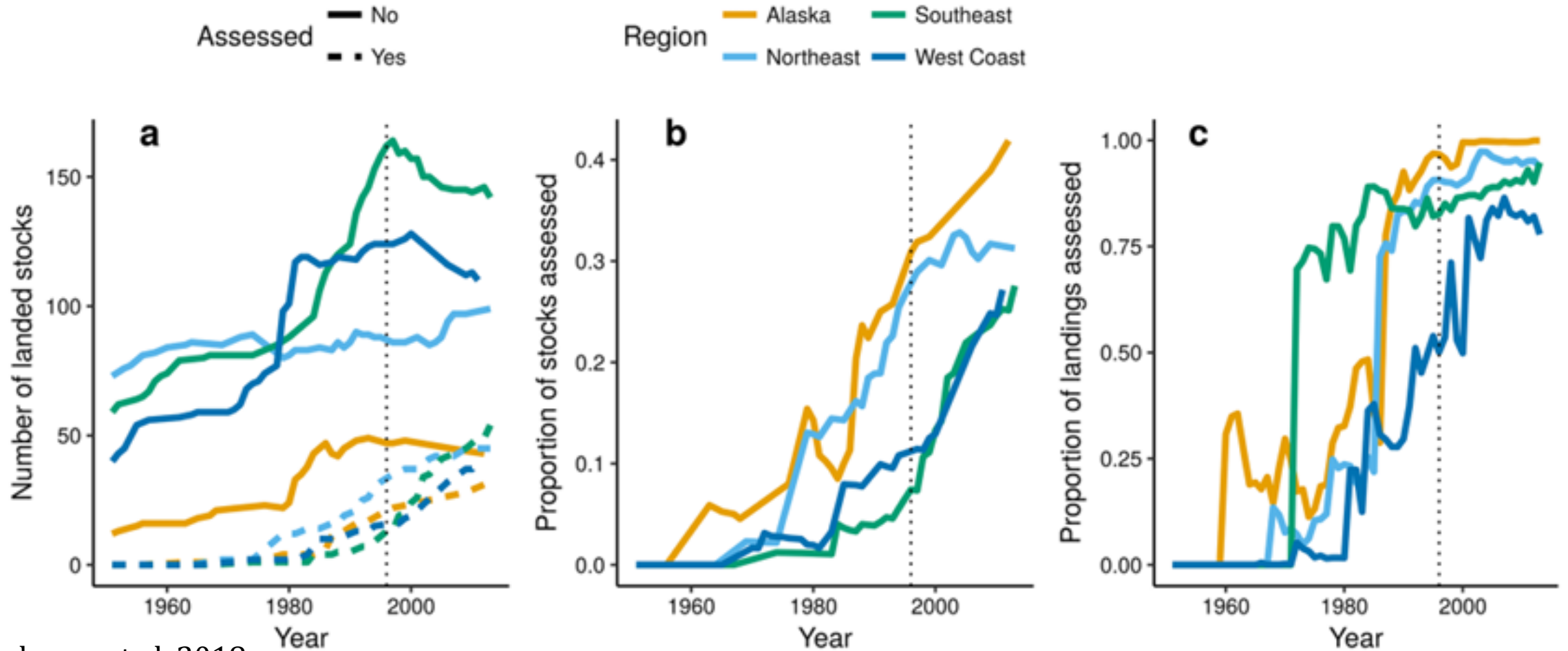
Increase in age structured stock assessments

Privitera-Johnson et al. 2020

- Age structured stock assessments increasing in number globally
- Leading to increase in demand for ages
- Agency capacity to produce age estimates has declined in recent year



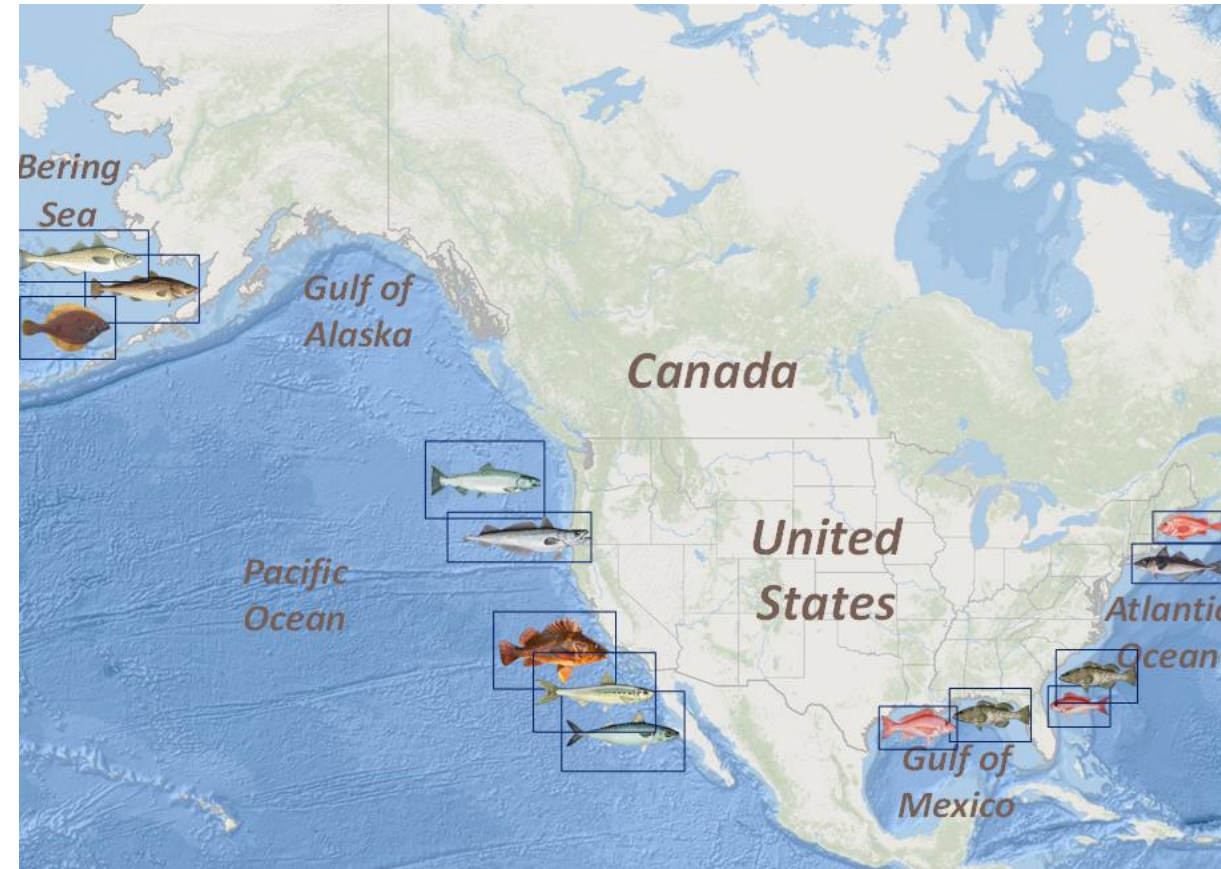
Increase in age structured stock assessments



Neubauer et al. 2018

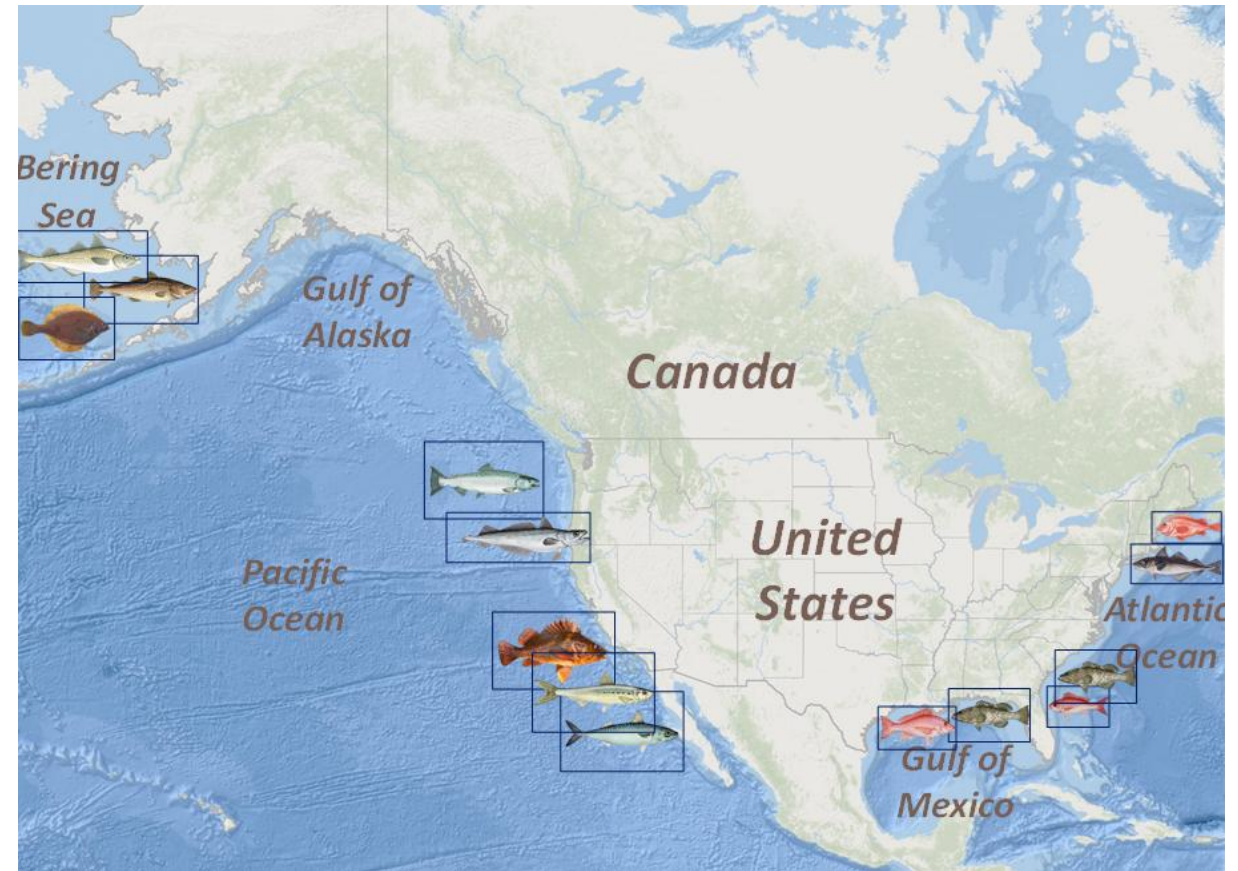
NOAA Fisheries FT-NIRS Strategic Initiative

- GOAL: Improve efficiency of age determination throughput while maintaining precision and with minimal effect on stock assessments
- Proof of concept work began in 2018 building on work from Australia
- SI proposal to NOAA Fisheries Science Board in 2019
- Funded SI 2020-2024, extended to 2025, one time supplemental in 2026



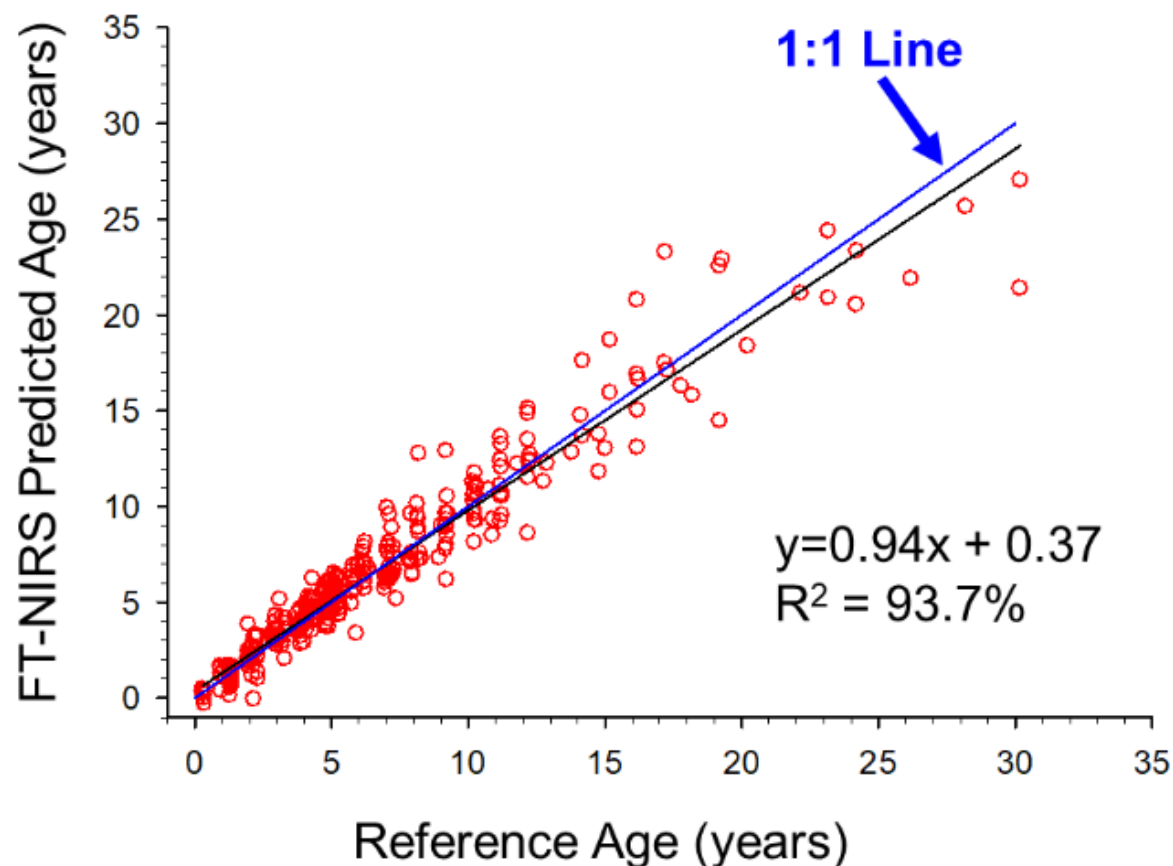
Research

- Each science center chose 3 species
 - Elasmobranchs
 - Daily ageing
 - Reproduction
 - Body condition
- Initial efforts used PLS models
- Mechanisms of FT-NIRS
- QA/QC
- Optimal Sample Design

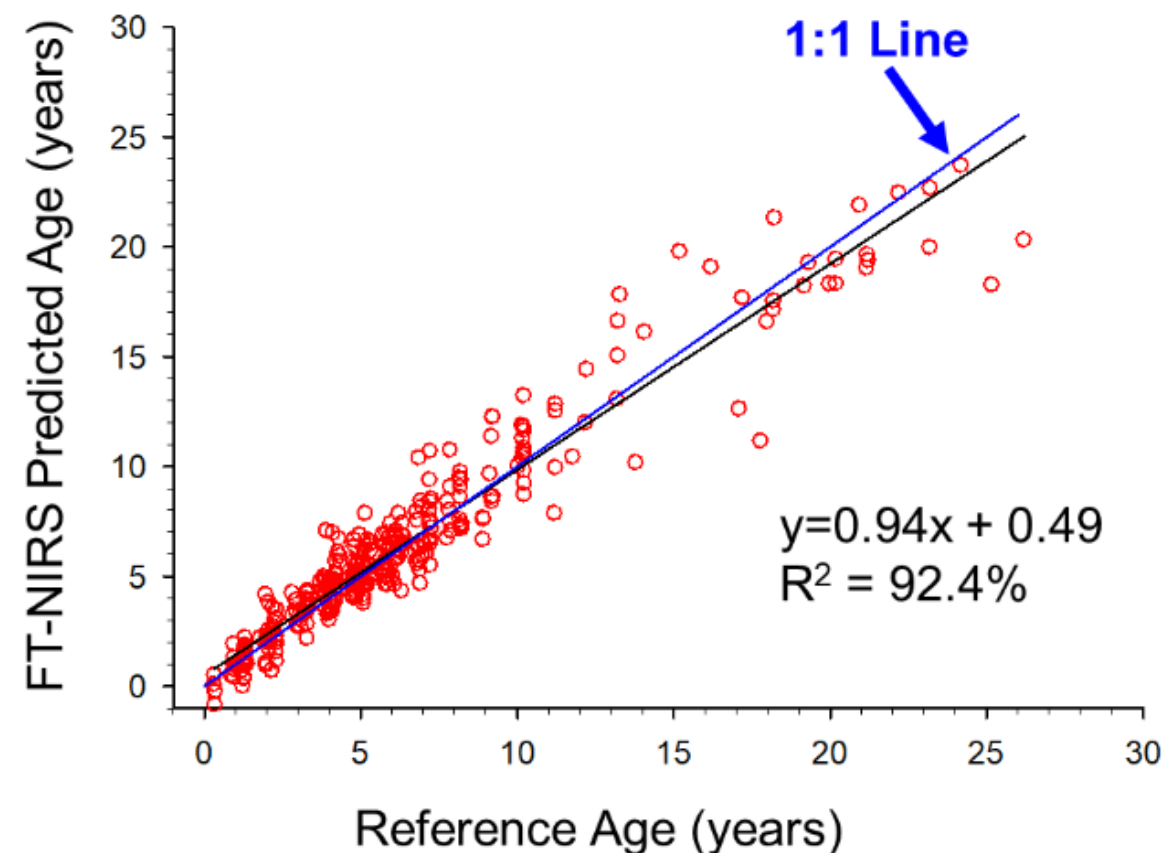


Research

Calibration: FT-NIR vs Reference Age



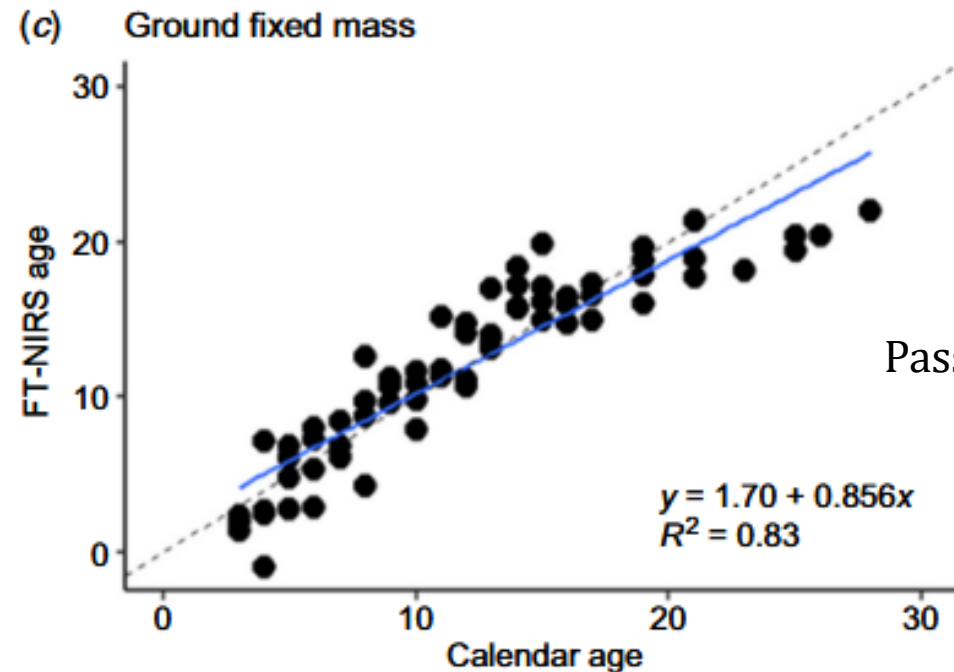
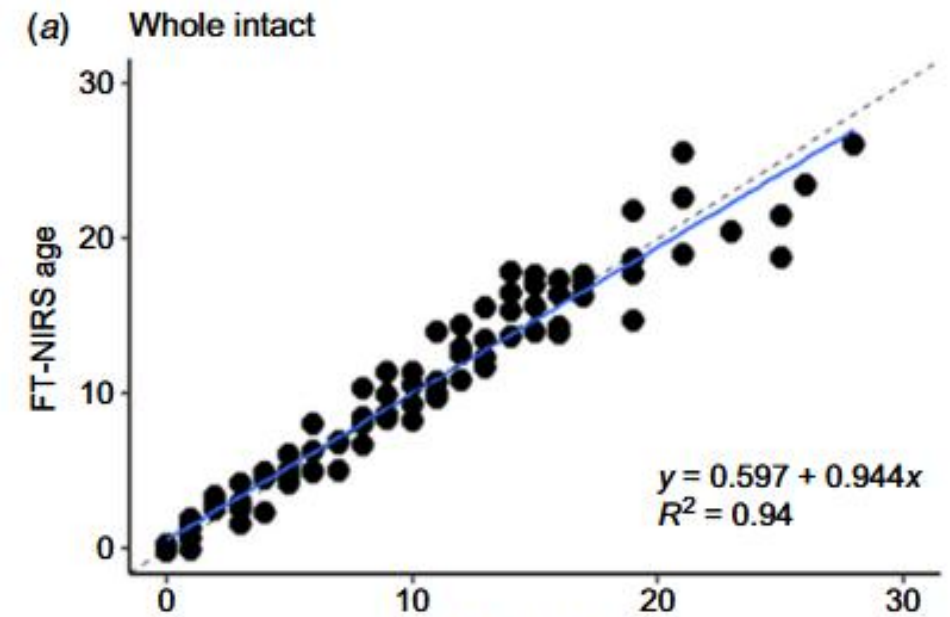
Validation: FT-NIR vs Reference Age



Barnett et al. 2019

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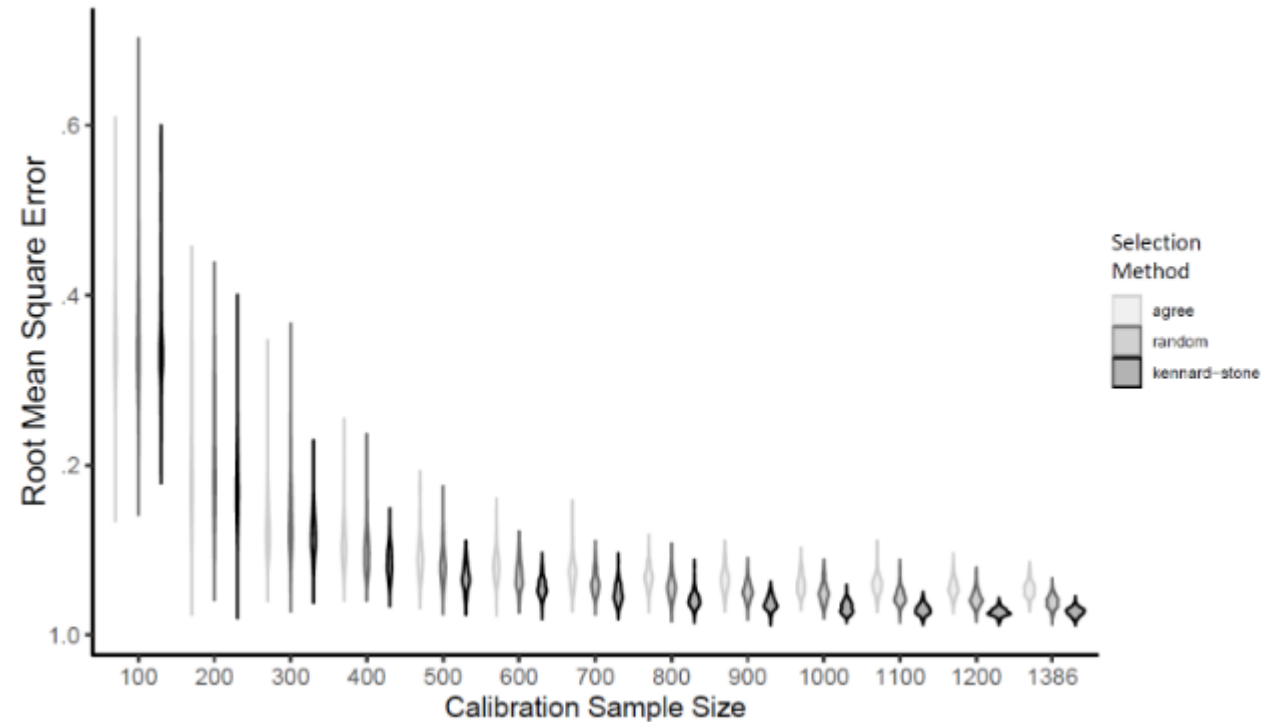
Passerotti et al. 2022



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Arrington et al. 2024

Original Article

Age estimation of red snapper (*Lutjanus campechanus*) using FT-NIR spectroscopy: feasibility of application to production ageing for management

Michelle S. Passerotti^{1*}, Thomas E. Helser², Irina M. Benson², Beverly K. Barnett³, Joseph C. Ballenger⁴, Walter J. Bubley⁴, Marcel J. M. Reichert⁴, and Joseph M. Quattro⁵

780



ARTICLE

A transformative approach to ageing fish otoliths using Fourier transform near infrared spectroscopy: a case study of eastern Bering Sea walleye pollock (*Gadus chalcogrammus*)

Thomas E. Helser, Irina Benson, Jason Erickson, Jordan Healy, Craig Kastle, and Jonathan A. Short



The future of fish age estimation: deep machine learning coupled with Fourier transform near-infrared spectroscopy of otoliths

Irina M. Benson^{1*}, Thomas E. Helser^{2*}, Giovanni Marchetti^{3*}, and Beverly K. Barnett^{4*}

Fish and Fisheries

SPECIAL ISSUE Fishing Into the Future: Using AI to Support Fish and Fisheries Research and Management

WILEY



ORIGINAL ARTICLE **OPEN ACCESS**

A Framework to Investigate the Effects of Observation Error on Neural Network Predictions of Fish Age

Derek W. Chamberlin¹, Thomas E. Helser², John D. Brogan³, Irina M. Benson⁴, Jason Conner⁵, Andrew T. Chin⁶, Christopher M. Gburski⁷, Mary Elizabeth Matta⁸, Julie A. Pearce⁹, Kalli R. Stone¹⁰, Todd T. TenBrink¹¹, Morgan B. Arrington¹²



Fourier-transform near infrared spectroscopy (FT-NIRS) rapidly and non-destructively predicts daily age and growth in otoliths of juvenile red snapper *Lutjanus campechanus* (Poey, 1860)

Michelle S. Passerotti^{a,*}, Christian M. Jones^b, Christopher E. Swanson^c, Joseph M. Quattro^a



RESEARCH PAPER
https://doi.org/10.1071/MF25053

MARINE & FRESHWATER RESEARCH

Advancing fish early life-history research: daily age determination of walleye pollock (*Gadus chalcogrammus*) with Fourier transform–near-infrared spectroscopy

Esther D. Goldstein^{A,*}, Mary E. Matta^A, Charles D. Waters^B, Heather K. Fulton-Bennett^B, Brenna C. Hsieh^A, Craig R. Kastle^A, Johanna J. Vollenweider^B, Jakub T. Sliwinski^C and Thomas E. Helser^A



RESEARCH PAPER
https://doi.org/10.1071/MF21341

MARINE & FRESHWATER RESEARCH

Physicochemical mechanisms of FT-NIRS age prediction in fish otoliths

Michelle S. Passerotti^{A,B,*}, Marcel J. M. Reichert^C, Bailey A. Robertory^{A,D}, Zachary Marsh^E, Morgan Stefik^E and Joseph M. Quattro^F

ECOSPHERE

COASTAL AND MARINE ECOLOGY

Aging Pacific cod (*Gadus macrocephalus*) from otoliths using Fourier-transformed near-infrared spectroscopy

JORDAN HEALY,¹ THOMAS E. HELSER,^{2,†} IRINA M. BENSON,² AND LUKE TORNABENE¹



Full length article

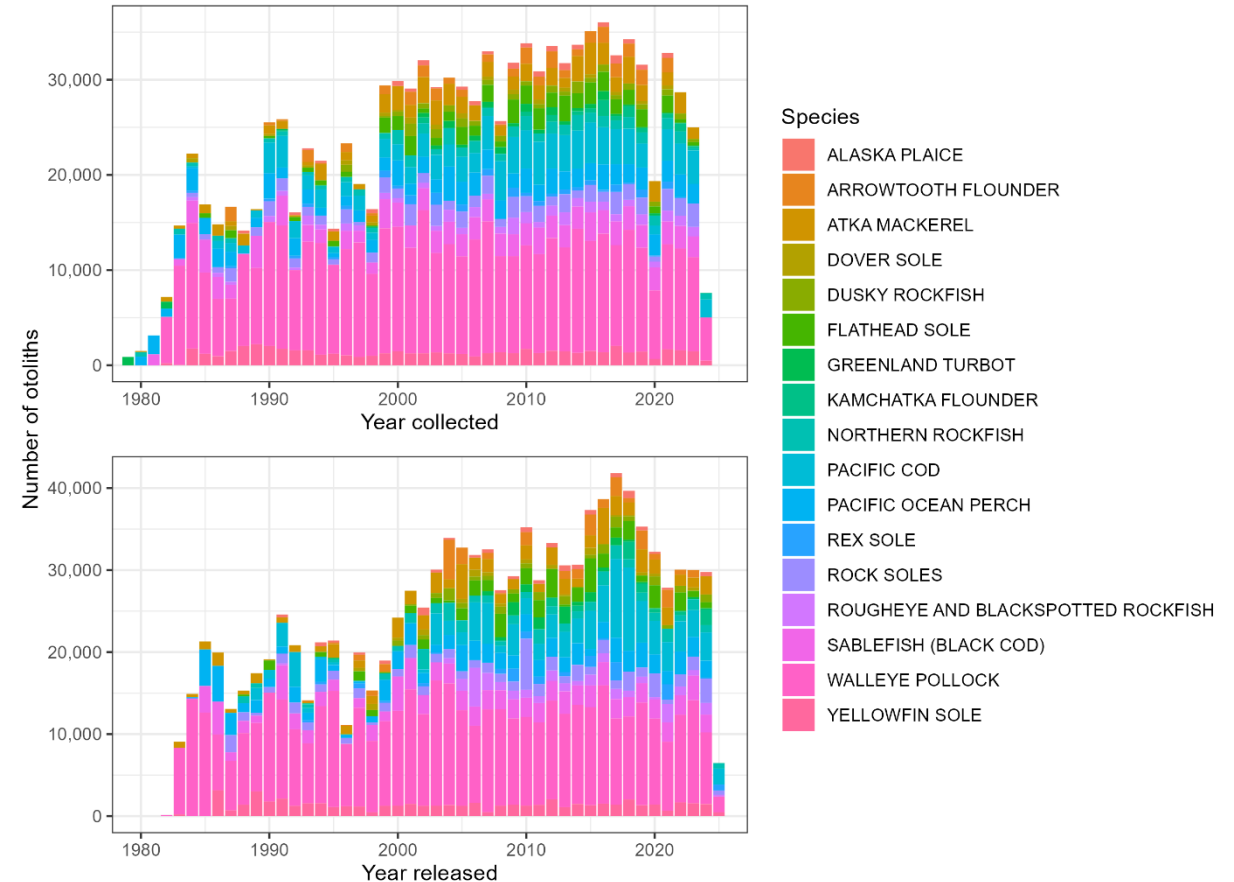
Fourier transform near infrared spectroscopy of otoliths coupled with deep learning improves age prediction for long-lived northern rockfish

Irina M. Benson^{a,*}, Thomas E. Helser^a, Beverly K. Barnett^b

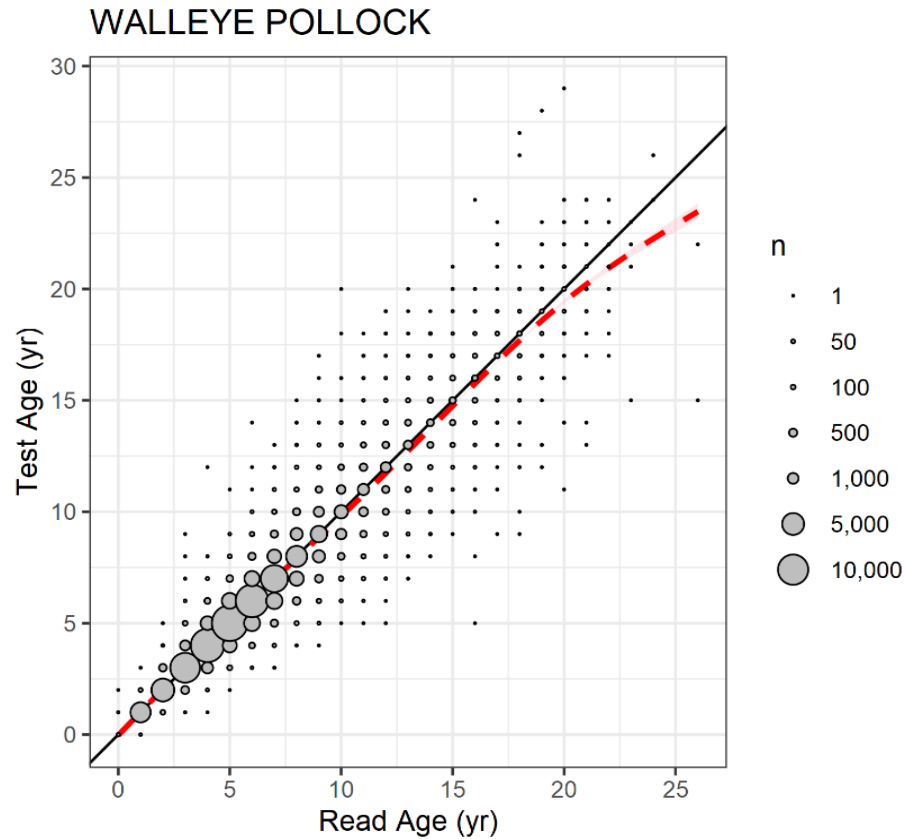


AFSC Age and Growth Program

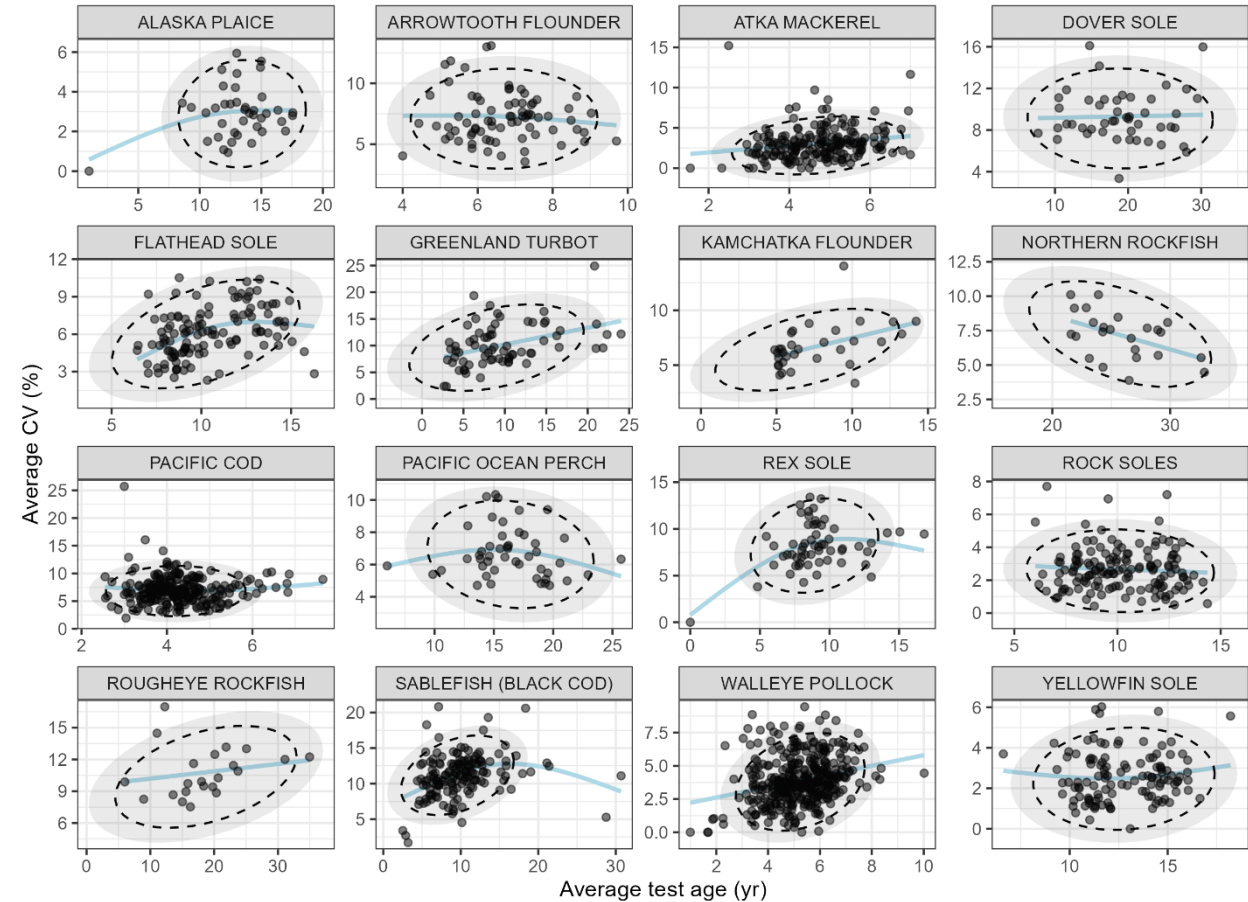
- 1 production team supervisor
- 5 production age readers
- 4 researchers
- 1 database manager
- 2 contractors
- Responsible for age determination of over 30 Alaska groundfish species
- 20 species aged on a production basis for use in age-structured stock assessments



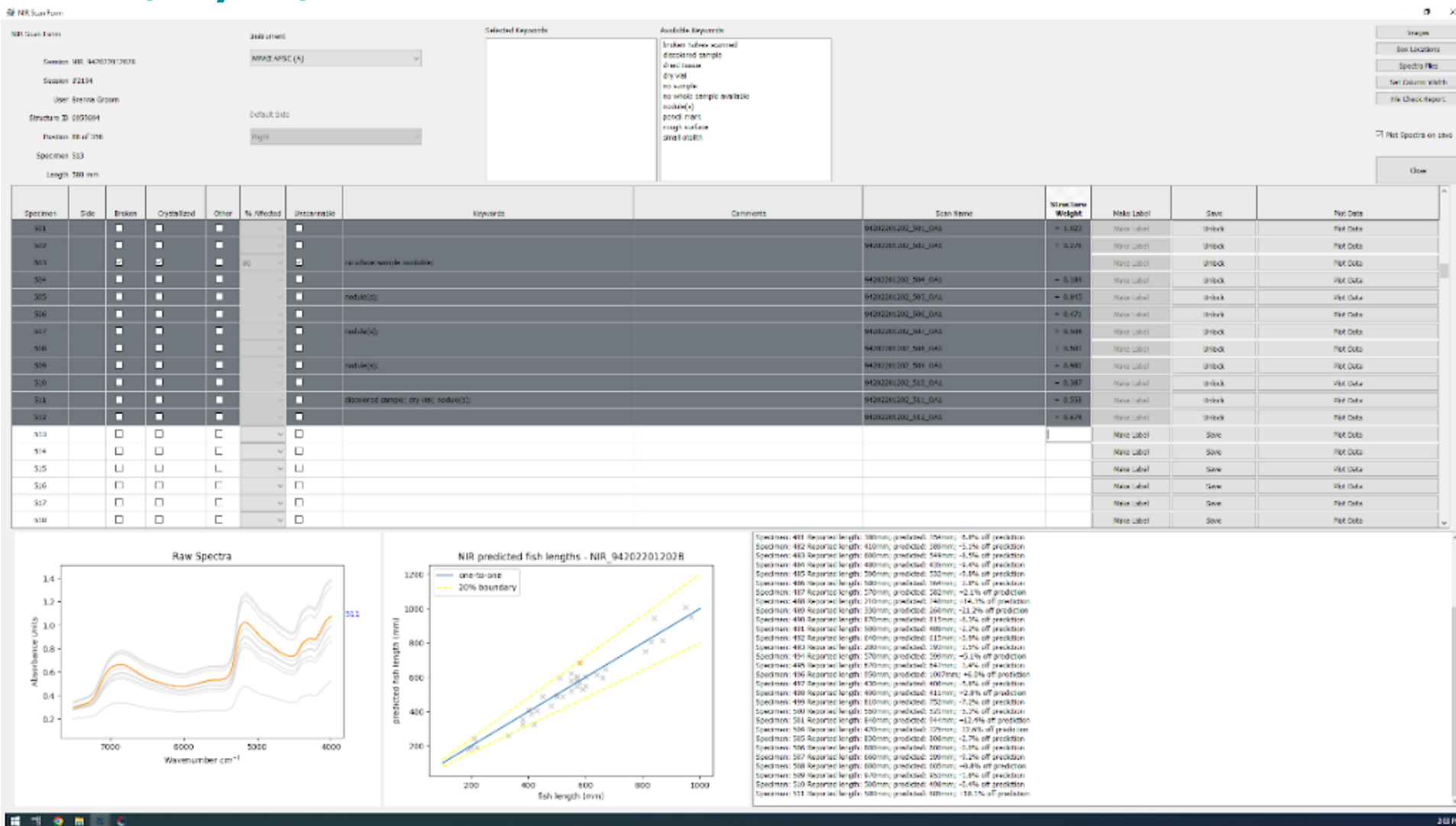
TMA Process and QA/QC



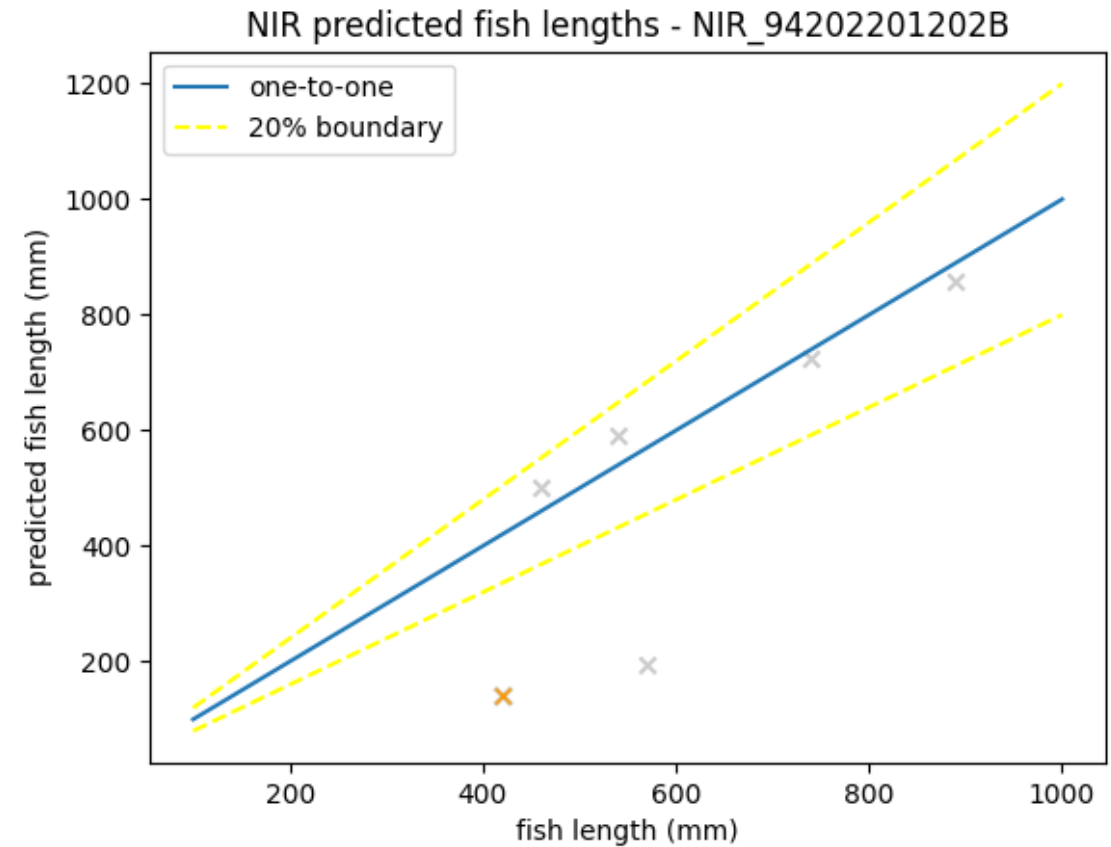
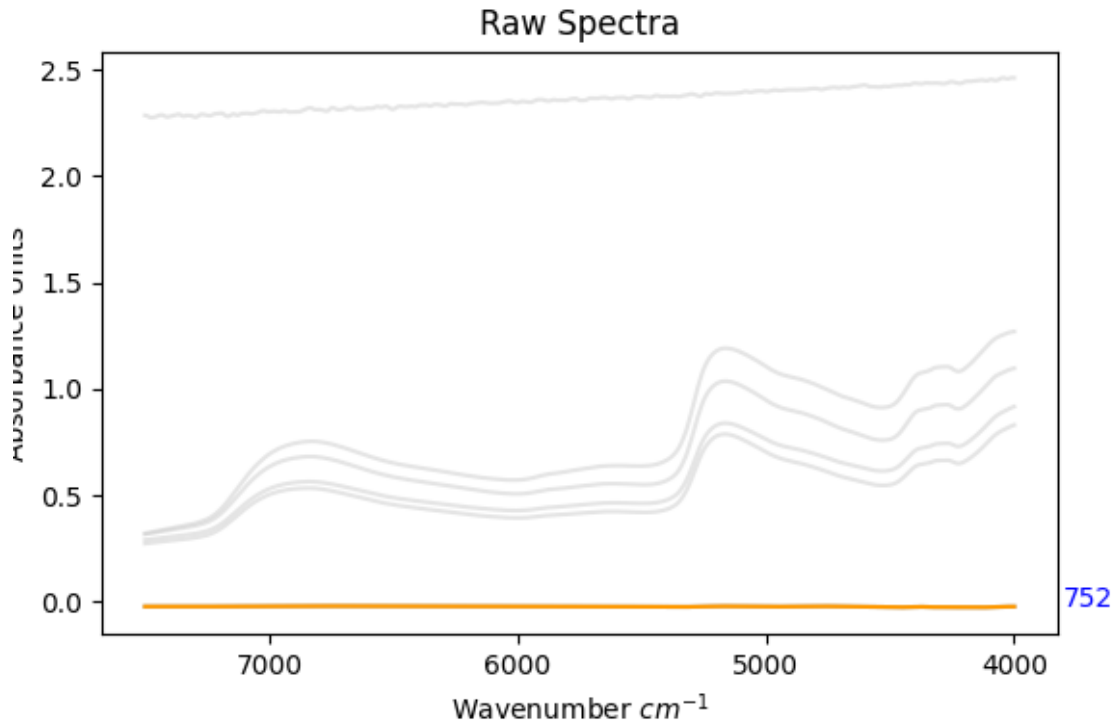
Matta et al. 2026



FT-NIR QA/QC



FT-NIR QA/QC



FT-NIR QA/QC

PCA and PLS outliers combined

- new spectra regular = gray
- new spectra PLS outlier = dark red
- new spectra PCA outlier = red

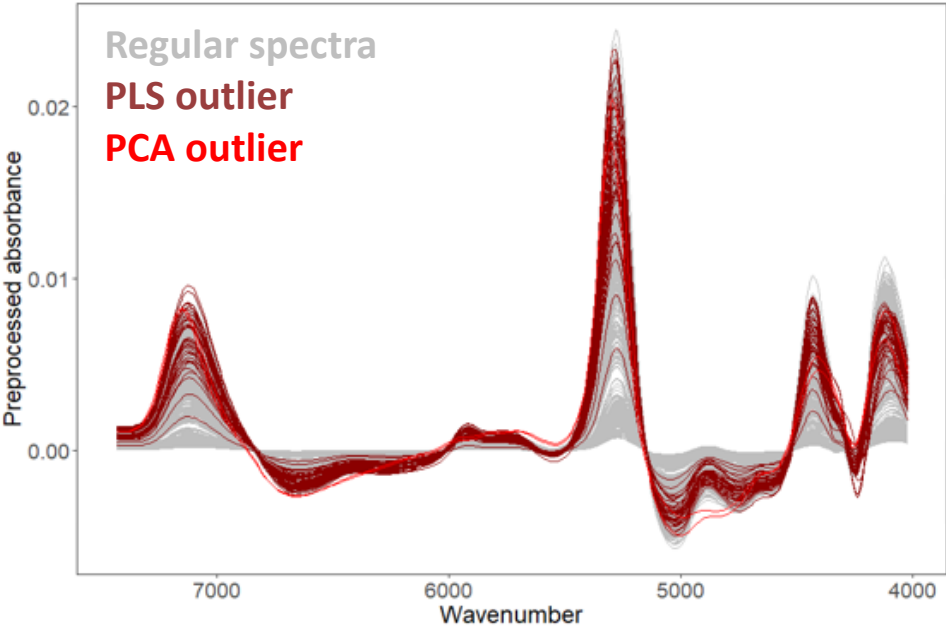


Figure 10. New data preprocessed spectra with PCA-based outliers in red and PLS-based outliers in dark red. Spectra that were not identified as outliers are in gray.

Goldstein et al. 2025

Table 4. Summary of usual spectra from PCA (Hotelling's T-squared and Q residuals) and PLS (Hotelling's T-squared, Q residuals, and Y-distances) using preprocessed data where p-values are based on a Chi-squared distribution. Critical limit extreme = 0.05 and critical limit outlier = 0.01

Total outliers	PCA outliers	PLS outliers
41	2	41

Table 5. List of file names for spectra identified as outliers from PCA and PLS approaches and information about potential issues with otoliths that may impact scan quality.

File name	Test PCA	Test PLS	broken	crystal-ized	other problem
WALLEYE_POLLOCK_162202201201_191_OA1.0	outlier	outlier			
WALLEYE_POLLOCK_94202201201_28_OA1.0	outlier	outlier			
WALLEYE_POLLOCK_162202201201_366_OA1.0		outlier		X	
WALLEYE_POLLOCK_162202201201_341_OA1.0		outlier			
WALLEYE_POLLOCK_162202201201_133_OA1.0		outlier		X	
WALLEYE_POLLOCK_162202201201_336_OA1.0		outlier			
WALLEYE_POLLOCK_162202201201_80_OA1.0		outlier		X	
WALLEYE_POLLOCK_162202201201_92_OA1.0		outlier		X	
WALLEYE_POLLOCK_162202201201_207_OA1.0		outlier			
WALLEYE_POLLOCK_162202201201_247_OA1.0		outlier	X	X	
WALLEYE_POLLOCK_162202201201_375_OA1.0		outlier		X	
WALLEYE_POLLOCK_162202201201_103_OA1.0		outlier		X	

FT-NIR QA/QC

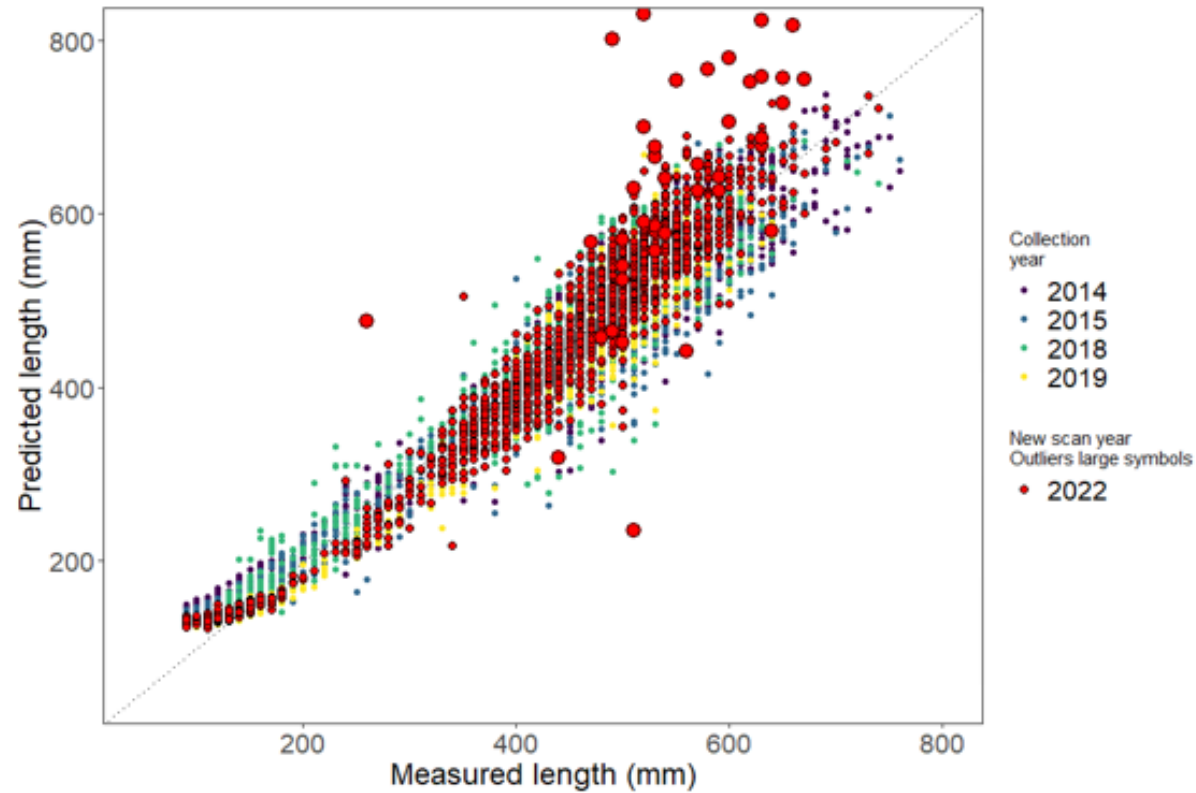


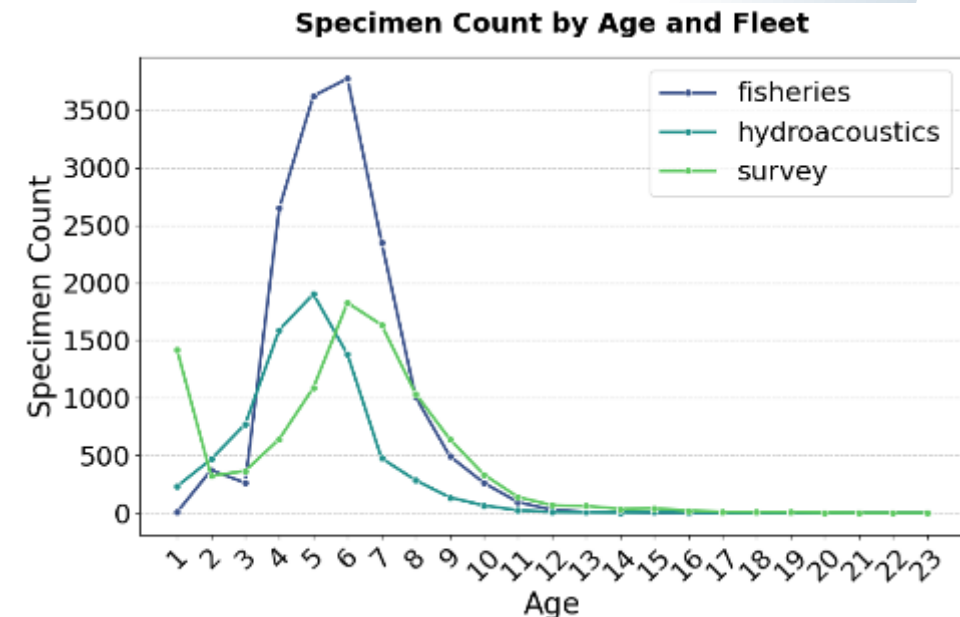
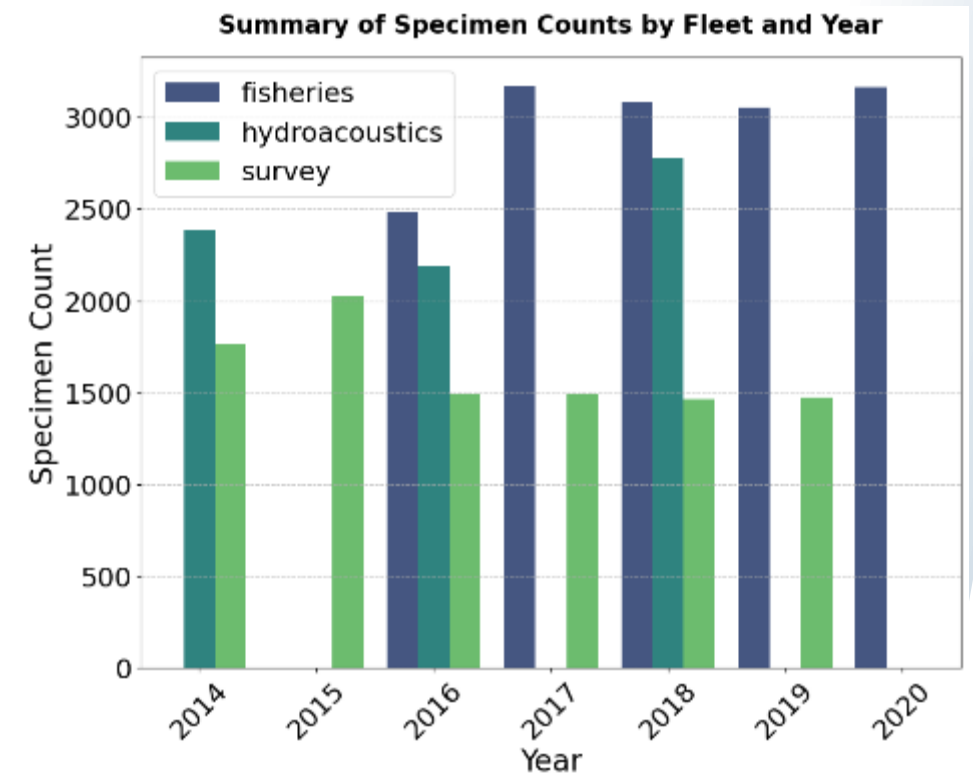
Figure 9. Observed and predicted fish length from PLS models including predictions from new data (New scan year). Large red points show outliers from new data based on Y-distance and full X-distance.

Goldstein et al. 2025

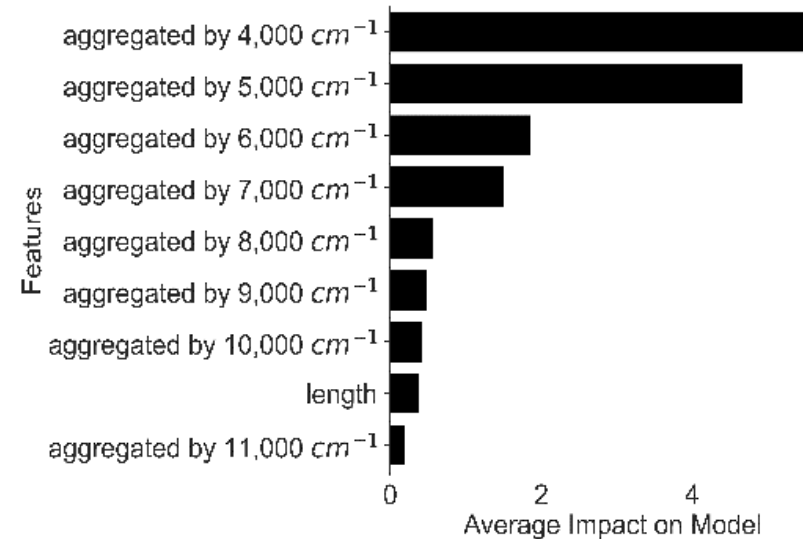
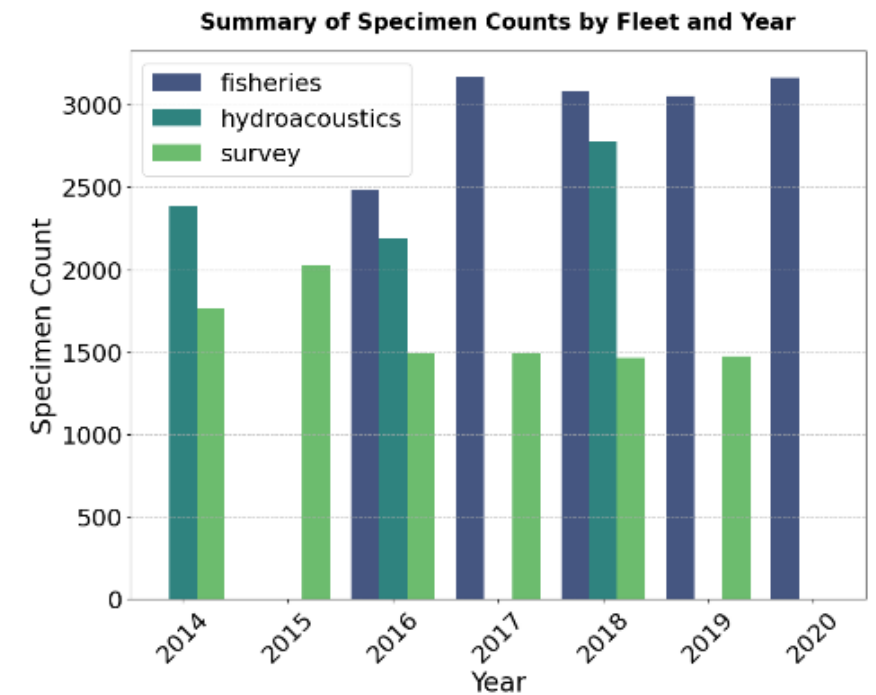
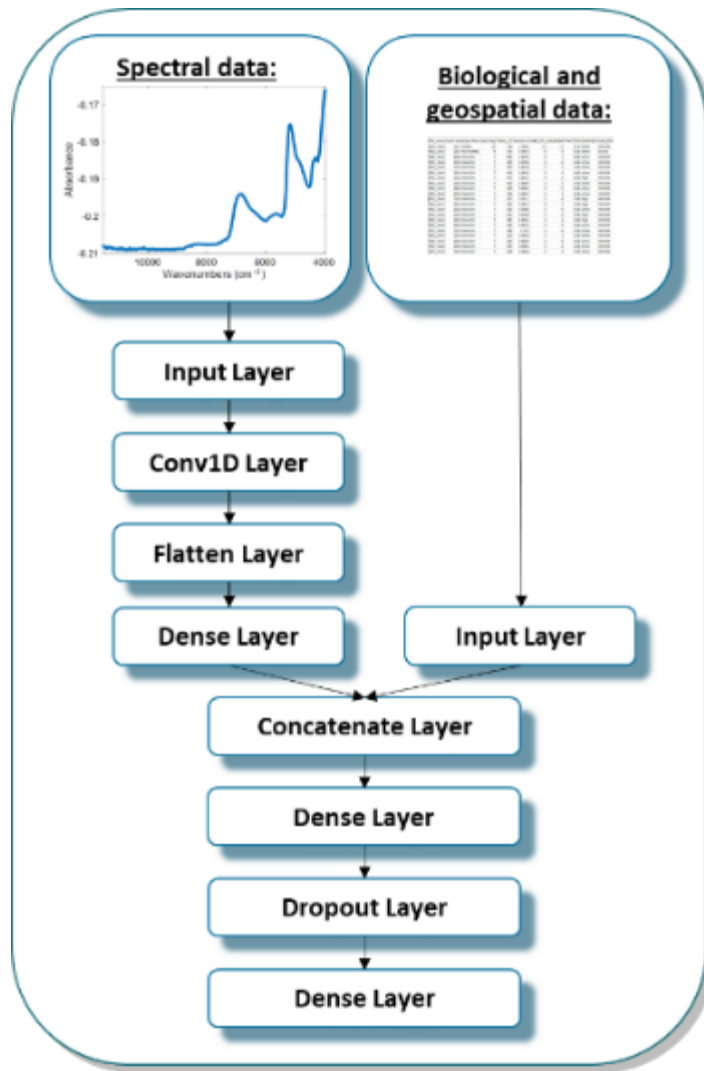
□

EBS Walleye Pollock Model

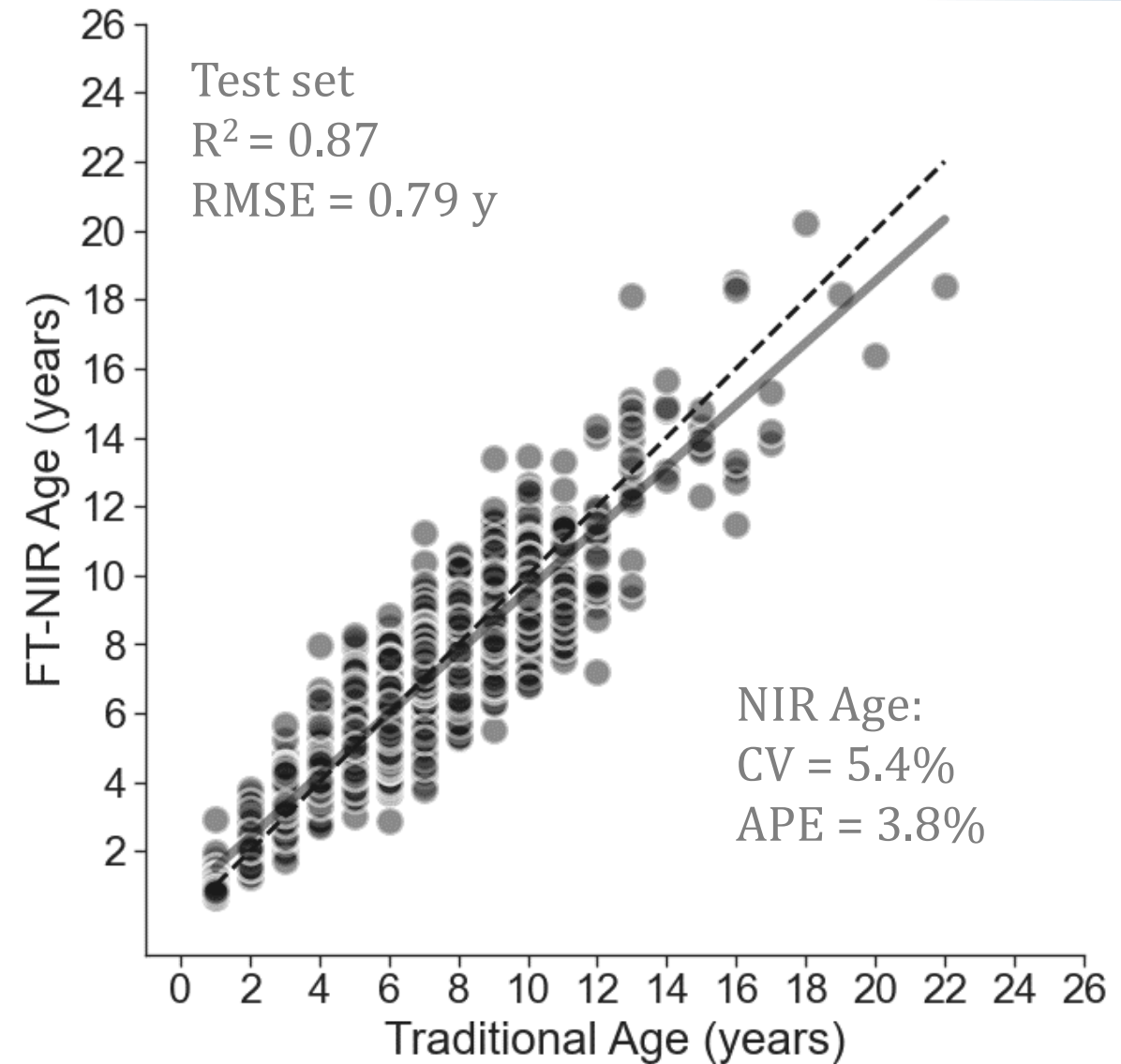
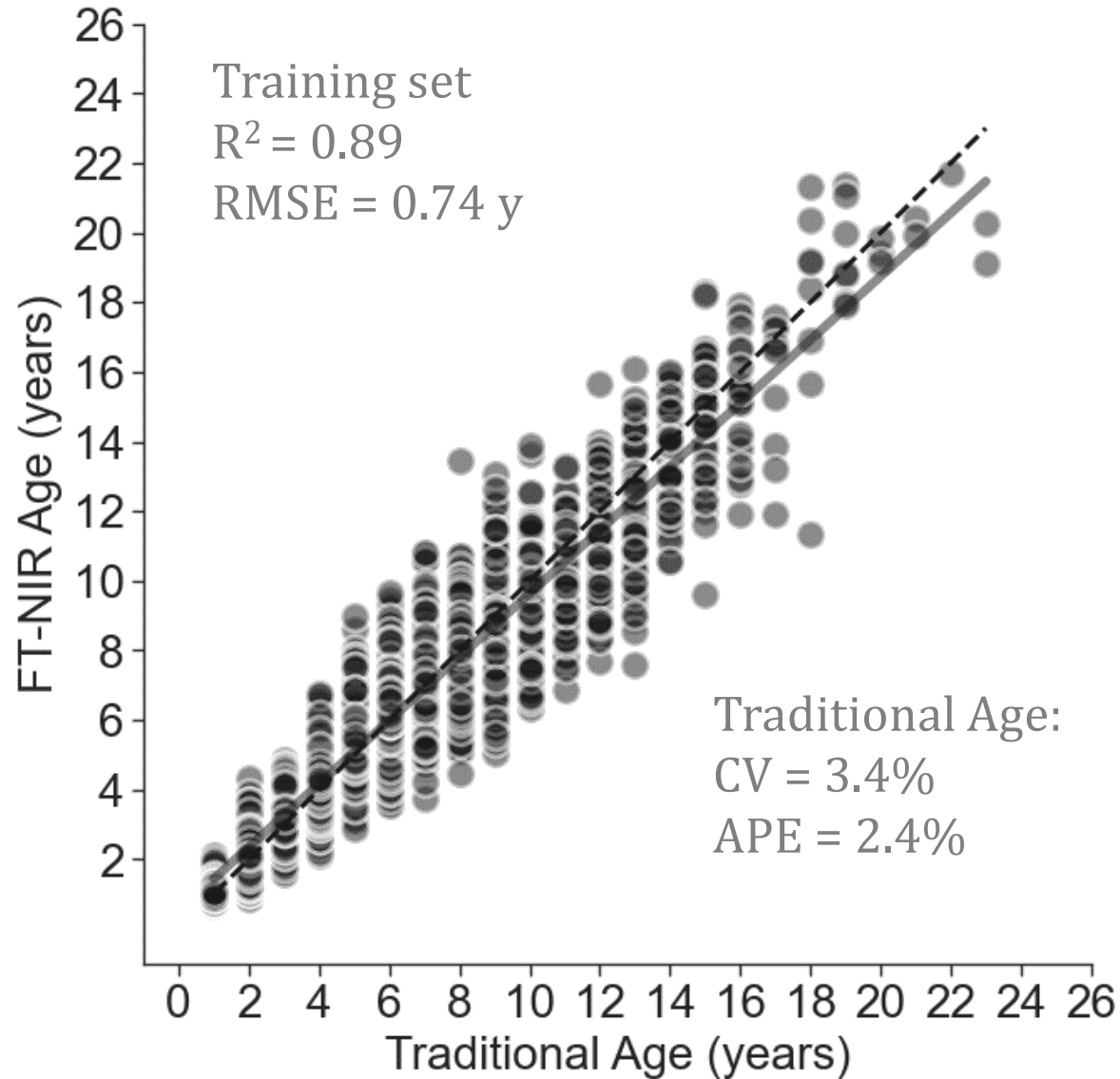
- Multimodal Convolutional Neural Network (MMCNN)
- Global (combined-fleet) model
- Training $n=25,625$
- Test $n=6,406$
- Stratified sampling (fleet and year)
- Custom loss function to give higher weights to rare age classes



FT-NIR MMCNN



EBS Walleye Pollock Model



FT-NIR Outliers

- PLS and PCA outliers TMA aged
- Monte Carlo prediction intervals from MMCNN
- If TMA and FT-NIR age falls outside the prediction intervals the sample is removed

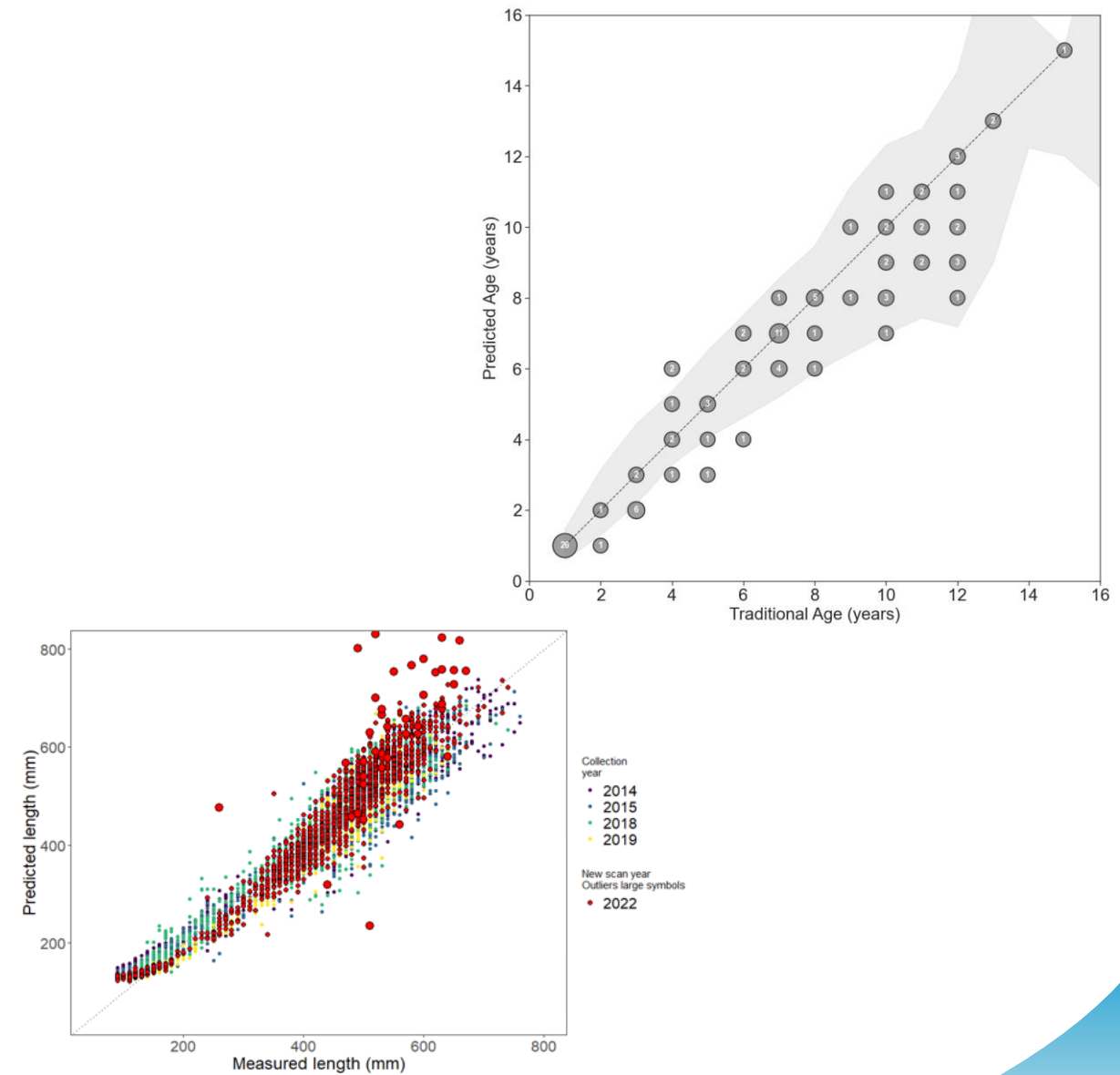


Figure 9. Observed and predicted fish length from PLS models including predictions from new data (New scan year). Large red points show outliers from new data based on Y-distance and full X-distance.

Future Plans

- Temporal decay of training data
- Machine learning post-doc at AFSC
 - Effect of sample selection on model performance
 - Classification vs. regression
 - R package
- Assessment integration and ageing error post-doc at SWFSC



Acknowledgments

- AFSC Age and Growth Team
- NOAA Fisheries FT-NIRS Strategic Initiative



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